Adaptive Multitask Autoencoder-Based Hyperspectral Unmixing Exploiting Auxiliary Data via Graph Associations

Jia Chen, *Member, IEEE*, Jun Li, *Fellow, IEEE*, and Paolo Gamba, *Fellow, IEEE*

***Abstract*— Hyperspectral unmixing is a technique in hyperspectral image processing that decomposes the spectra of mixed pixels into pure spectral components (endmembers) and their corresponding contributions (abundances). When dealing with complex mixed-terrain scenes, such as urban areas, significant challenges arise due to the complexity of the environment. Urban areas feature intricate geometric structures in individual pixels, including diverse 2-D and 3-D structures and the composite use of various building materials, resulting in highly complex scenarios. To address these challenges, this work exploits urban auxiliary information in the framework of an adaptive multitask autoencoder (AE) unmixing model, utilizing graph associations. The framework enhances the information in hyperspectral images by utilizing urban auxiliary data. Specifically, it performs superpixel segmentation to subdivide complex urban environments into simpler units. Subsequently, different AE-based unmixing methods are applied to these segmented results. Graph associations are employed to identify similar blocks in the image, incorporating this additional information into the unmixing process. In the experiments conducted for this work, two hyperspectral unmixing datasets were prepared, along with their corresponding urban auxiliary data. The results demonstrate that the proposed method achieves robust performance, even in complex urban environments.**

***Index Terms*— Autoencoders (AEs), graph associations, hyperspectral imaging, multitask learning, remote sensing, superpixel segmentation, unmixing.**

I. Introduction

HYPERSPECTRAL remote sensing data has become increasingly popular due to its ability to provide nearly continuous spectral information. This capability makes it a powerful tool for detecting and identifying surface covers [1], [2]. However, the high spectral resolution of hyperspectral data is offset by a relatively coarse spatial resolution. This often results in mixed pixels containing multiple materials [3], [4], which complicates the identification of these materials using solely spectral information. To overcome this challenge, researchers have developed spectral unmixing (SU) techniques. These techniques decompose mixed pixels into their pure spectral signatures and associated abundances. SU has been extensively applied in hyperspectral image analysis, demonstrating significant potential in enhancing material identification and mapping accuracy.

Due to the prevalence of the linear mixture model (LMM) assumption, most SU methods are based on it and implement linear unmixing techniques [5], [6], [7]. The LMM posits that sunlight is reflected only once off the surface and that the observed spectrum is a linear combination of the endmembers. This assumption allows for the development of various linear unmixing algorithms, which estimate the abundance fractions of each endmember in a mixed pixel.

Linear unmixing [8], [9], [10], [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21] has been extensively used in hyperspectral image analysis due to its simplicity and effectiveness. It finds applications in land-cover classification, mineral mapping, and environmental monitoring, efficiently handling large datasets and being easily implementable in software packages.

There are several common linear unmixing methods used in hyperspectral image analysis, subdivided into the following.

1) Geometry-based methods, such as the pixel purity index (PPI) [8], N-FINDR [9], and vertex component analysis (VCA) [10].
2) Nonnegative matrix factorization (NMF)-based methods, like robust collaborative NMF [11], MVCNMF [13], $L_{1/2}$-NMF [14], and sparsity-constrained deep NMF [15].
3) Bayesian-based methods, including hierarchical Bayesian [22] and joint Bayesian [18].
4) Sparse-based methods, such as matching pursuit [16] and RGSU [17].
5) Evolutionary computation-based methods, for example, EE-AOC [23] and TSEA [24].
6) Autoencoder (AE)-based unmixing methods, like UDAS [19] and DAEN [20].

However, an important limitation of linear unmixing is its potential inaccuracy in real-world scenarios. To address this issue, researchers have proposed various nonlinear unmixing algorithms, including bilinear models and deep learning-based models.

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Bilinear unmixing [25], [26], [27], [28], [29], [30] is an SU technique that handles nonlinear mixing models by assuming the observed spectrum to be a bilinear combination of endmembers. This method has proved effective in modeling complex mixing scenarios, such as those in urban environments and agricultural fields.

In bilinear unmixing, the observed spectrum is modeled as a bilinear combination of two sets of endmembers, each representing a different physical material. The abundance fractions of each endmember set are estimated separately, using optimization techniques like alternating least-squares (ALS) or NMF [31]. Semi-NMF [25] introduces a novel optimization method for nonlinear unmixing based on the generalized bilinear model (GBM), optimizing the processing of an entire image in matrix form. Additionally, robust NMF (rNMF) [26] proposes a robust mixing model for describing hyperspectral data resulting from material mixtures. Furthermore, kernel-based nonlinear unmixing methods have also been developed. For example, [27] suggests using kernel archetypal analysis (KAA) for flexible endmember extraction, which accounts for intraclass variability in relative pure pixel mapping and mixed pixel unmixing. In the realm of AE, deep multitask bilinear unmixing (DMBU) [28] offers an unsupervised unmixing approach based on a hierarchical bi-LMM. This approach addresses two different tasks in a multitask framework and has shown promising results in hyperspectral image analysis.

Several higher-order nonlinear unmixing methods have been proposed to address more complex scenarios involving high-dimensional data, non-Gaussian noise, and unknown mixing models. These methods typically incorporate sophisticated mathematical models and optimization algorithms that capture the nonlinear and nonconvex nature of the unmixing problem. However, they often demand significant computational resources and a deep understanding of tensor algebra, optimization theory, and machine learning. Consequently, their practical application remains limited to specific domains or research areas where high accuracy and interpretability are paramount.

All these methods encounter limitations in certain scenarios, as they strive to balance accuracy, computational efficiency, and interpretability. For instance, city administrators require timely responses to environmental threats and could benefit greatly from the rich spectral data provided by hyperspectral sensors to detect air pollutants, unhealthy vegetation, and hazardous materials. However, analyzing urban environments using hyperspectral datasets is complex due to the prevalence of mixed pixels. These pixels often contain multiple endmembers, especially in urban areas, where the intermingling of roads, buildings of different materials, and vegetation is common. In spaceborne hyperspectral images of urban areas, with their typical spatial resolution of tens of meters, mixed pixels with multiple endmembers and varied reflection conditions are very common.

This article, therefore, proposes an adaptive multitask AE-based model that leverages graph associations to tackle the aforementioned challenges in urban areas. The approach integrates additional auxiliary information to enhance the interpretation of hyperspectral images. This auxiliary data guides the superpixel segmentation of the hyperspectral image, with each segmented superpixel being assigned a specific unmixing method. In this manner, multiple methods can be applied concurrently, significantly reducing overall computational time. Furthermore, to extract more information from the hyperspectral image, the algorithm employs graph associations to associate similar blocks in the image, thereby reducing noise interference.

In summary, the key contributions of this work are as follows.

1) A multitask parallel processing unmixing method that significantly reduces time consumption.
2) A graph association approach for correlating similar blocks in hyperspectral images, which is highly effective in mining additional information and achieving information gain in hyperspectral image analysis.

These contributions underscore the effectiveness and applicability of the proposed method in hyperspectral data unmixing, particularly in scenarios where the complexity is high and a singular unmixing model is not suitable for all parts of the scene. The quantitative analysis, conducted using two new datasets, also enriches the pool of openly available datasets for evaluating hyperspectral processing techniques.

The introduction of research related to autoencoding SU is delineated in Section II. The proposed method is presented in detail in Section III. The experiments and results are shown in Section IV. Eventually, the conclusion is drawn in Section V.

II. Related Work

A. AE-Based SU

In the realm of hyperspectral unmixing, the application of AEs plays a pivotal role in breaking down complex spectral mixtures into their fundamental spectral components or endmembers. An AE is composed of two main parts: an encoder and a decoder. The purpose of this structure is to capture a reduced, low-dimensional representation of the input data. The encoder function compresses the original, high-dimensional data, mapping it onto a latent space. This process is then inverted by the decoder, which projects the data from the latent space back to its original high-dimensional form. Throughout the training phase, the network is meticulously fine-tuned to minimize the disparity between the original input and its reconstructed output.

Considering that the fundamental mechanism of AEs aligns closely with that of SU, their deployment in hyperspectral unmixing has been demonstrated to be both precise and computationally efficient. This approach stands out favorably when juxtaposed with other cutting-edge technologies.

1) Linear AE-Based Unmixing: The method of linear AE-based unmixing utilizes a linear AE to distill low-dimensional features from high-dimensional input data. The network parameters are then further refined, aiming to reduce the reconstruction error (RE) between the original data and its reconstructed counterpart.

Various methodologies have been adopted to augment this base framework. For instance, Licciardi and Del Frate [32]

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employed a multilayered AE architecture for hyperspectral unmixing, while [19] l_{21}-norm in the encoder to simultaneously minimize RE and eliminate redundant endmembers has been considered. Other innovations include denoising-oriented unmixing [33], which exploits the capabilities of AE to address unmixing challenges, and algorithms that blend automated sampling with nonnegative sparse autoencoding to handle outlier issues [34]. Specifically, the work in [34] imposes a 1-norm to the latent output to generate sparse unmixing results. In [35], stacked nonnegative sparse AEs are utilized to handle data affected by outliers and a low signal-to-noise ratio. Researchers have also explored the use of convolutional neural network AE (CNNAEU) [36], an architecture that considers both spatial and spectral structures, as well as deep networks [37] based on global–local smoothing AEs. Jin et al. [38] proposed a novel technique network for unsupervised unmixing which is based on adversarial AEs, while Shahid and Schizas [39] proposed a novel AE structure combined with a kernelization layer, mapping the mixed pixels in a higher-dimensional space for easier separability along with a novel cross-product layer to account for nonlinear mixing mechanisms. In [40], an AE network with adaptive abundance smoothing has been presented. Various approaches have been proposed to enhance unmixing performance by exploiting spatial–spectral correlations through regularization techniques. Semi-NMF-RE [41] introduced a proximal regularization approach that guides the trained AE to capture the characteristics of the linear mixture generative model inherent in hyperspectral imagery. AERED [42] integrates AE networks with regularization by denoising through a two-step optimization strategy, combining the implicit learning capability of deep learning with explicit prior information from denoisers to enhance unmixing interpretability and performance.

The primary limitation of the linear AE-based unmixing method is deeply rooted in its core assumption that the spectra observed are linear combinations of endmembers. While this simplifying assumption is useful in less complex environments with minimal mixing and only single reflection conditions, it poses considerable challenges in more intricate scenarios. In complex urban landscapes where multiple layers of materials often overlap and interact, resulting in high levels of mixing, multiple reflections, and even nonlinear interactions, the linear AE approach struggles to deliver accurate and reliable unmixing results. Not only does the method find it difficult to precisely identify the types of materials present but it also faces challenges in determining their respective proportions or abundances within each pixel. Furthermore, because the method is based on linear assumptions, it is not well-equipped to handle nonlinearities that frequently occur in real-world applications. Consequently, its applicability becomes severely restricted when encountering multifaceted, heterogeneous environments. This makes it imperative to consider alternative or supplementary techniques when dealing with complex scenarios, particularly if the aim is to achieve high-precision, reliable hyperspectral unmixing.

2) Bilinear AE-Based Unmixing: As a sophisticated technique in hyperspectral unmixing, bilinear AE-based unmixing capitalizes on the strengths of bilinear models to accurately depict the concentrations of diverse materials in individual pixels. This approach marks a departure from conventional linear models, employing a bilinear AE for the parameter learning process, thus offering a more detailed perspective on the distribution of materials. The utilization of bilinear models allows this method to adeptly handle the more intricate interrelations among materials, their abundances, and the captured spectra. This capability not only enhances the method's potential in accurately mapping material distributions but also bolsters the overall reliability of the unmixing process.

Zhan et al. [43] presented a ground-breaking deep AE network inspired by the GBM, specifically developed to address the complexities of nonlinear unmixing. This architecture smartly incorporates regularization terms to minimize the risk of overfitting, leveraging existing knowledge in the problem domain. In contrast, the DMBU model [28] introduces an innovative unsupervised technique rooted in a hierarchical bilinear mixture framework. This approach makes strategic use of hierarchical structures, enabling a detailed analysis of the complex interrelationships among various elements in hyperspectral data.

The bilinear AE-based unmixing method, while effective as a nonlinear technique for handling complex mixed pixel scenarios and high levels of mixing, brings its own set of challenges. A key tradeoff lies between computational speed and accuracy. Due to its dependence on complex computations and detailed parameter tuning, this method can be computationally intensive. Additionally, there is a notable risk of model overfitting, a concern that requires vigilant management throughout the implementation process.

3) Nonlinear AE-Based Unmixing: Nonlinear AE-based unmixing [44] represents a method in hyperspectral unmixing that employs nonlinear AEs to derive a low-dimensional representation of the input data. This method optimizes network parameters by simultaneously minimizing RE and incorporating regularization terms. Wang et al. [45] introduced an innovative approach for nonlinear unmixing using a sophisticated deep AE network. Additionally, LSTM-DNN [46] presents a unique nonlinear unmixing scheme leveraging the capabilities of long short-term memory networks and deep neural networks to capture the sequential attributes of hyperspectral pixels.

Shahid and Schizas [47] detailed a fully unsupervised, AE-based approach for nonlinear SU, employing a novel averaging method with spatially aware filters created through radial basis function (RBF) kernels. The work in [48] proposes a residual dense AE network (RDAE) tailored for nonlinear hyperspectral unmixing in scenarios involving multiple scattering. Moreover, 3-D convolutional neural networks (3DCNNs) [49] emphasize the enhancement of spatial–spectral correlation, thus improving hyperspectral unmixing results. EMLM-Net [50] proposes an extended multilinear mixing model-inspired dual-stream network for unsupervised nonlinear unmixing. MLM-AE [51] proposes an AE-based framework that explicitly models multilinear mixing through a Markov chain perspective, offering both pixel-wise and spectral–spatial solutions to capture infinite-order endmember interactions in hyperspectral unmixing. Lastly, the multitask

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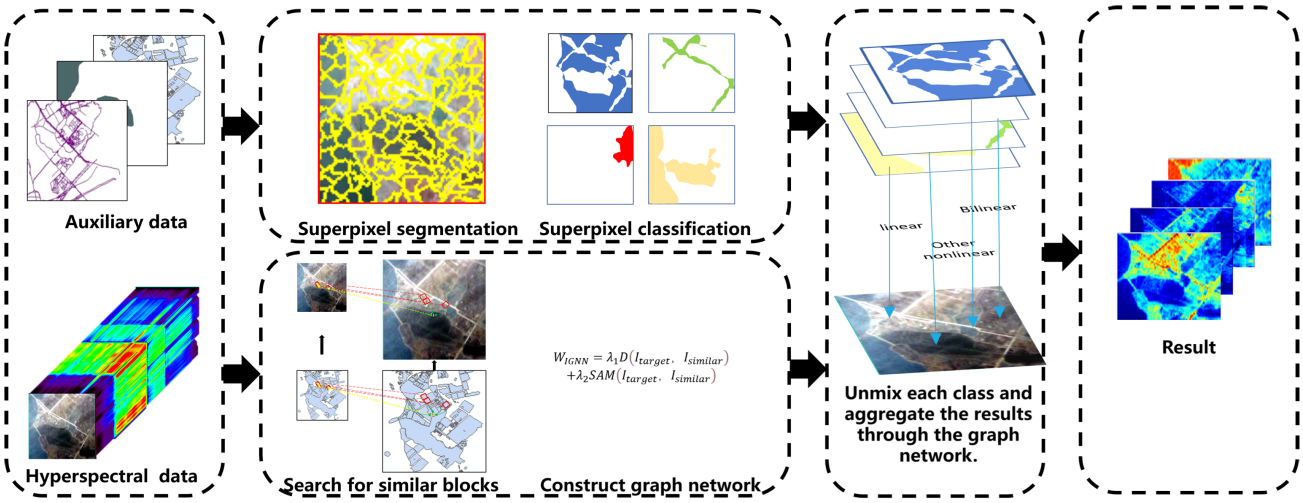


Fig. 1. Architecture of our proposed framework.

autoencoder hyperspectral unmixing model (MAHUM), [52] offers an adaptive unmixing approach that tailors the unmixing model to the reflectance characteristics of pixels, significantly enhancing unmixing precision.

III. Methodology

The main contribution of this article lies in exploiting spatial–spectral correlations through auxiliary data in a multitask unmixing framework. Specifically, we construct spatial–spectral relationships in two aspects: 1) utilizing auxiliary data to guide the allocation of appropriate unmixing models to different scene subareas, which not only enhances unmixing accuracy but also ensures computational efficiency by avoiding the unnecessary application of complex models where simpler ones suffice and 2) leveraging graph associations to capture intrinsic spatial patterns and similarities among image blocks, which strengthens the spatial–spectral correlation modeling in the unmixing process. This dual-path strategy effectively integrates auxiliary information and spatial context to improve unmixing performance.

To better introduce its contents, this section will be divided into three parts. As illustrated in Fig. 1, our framework consists of three main components. In Section III-A, the task to implement a multitask unmixing model through auxiliary information will be described in detail. Among them, there are two parts: superpixel segmentation and superpixel classification. How to find similar blocks and build graph associations will be introduced in Section III-B, while how to obtain results through a multitasking method and aggregate the results through graph associations will be introduced in Section III-C. The overall flow of the algorithm is shown in Algorithm 1.

A. Multitask Unmixing Model Based on Superpixel Segmentation

In hyperspectral image analysis, superpixel segmentation is an effective method for reducing the data's dimensionality and extracting meaningful features. By dividing the image into several superpixel blocks, each block can be treated as a distinct entity for further analysis. In this work, we apply the simple linear iterative clustering (SLIC) [53] algorithm to perform superpixel segmentation, resulting in several classes of superpixels. A different label is assigned to each class based on its spectral characteristics and spatial positions. This process aids in simplifying the complexity of hyperspectral data and facilitates subsequent analysis tasks such as classification and target detection. The following is a detailed description that delves into the inner workings and various applications of the SLIC algorithm.

1) The SLIC algorithm converts the image to the laboratory color space and then divides it into equally sized grids. For each grid, the center point is selected as the initial clustering center, and the pixel with the smallest color distance to that center point within a $2S \times 2S$ neighborhood centered on that point is searched as the clustering center. S is a preset parameter used to control neighborhood size. For different datasets, we start with a default value commonly used in literature and then perform a grid search to find the optimal value by evaluating the segmentation quality on a validation set. The optimal S is chosen based on the highest segmentation accuracy or the best balance between over-segmentation and under-segmentation.
2) For each clustering center, all pixels within a $2S \times 2S$ neighborhood centered on it are assigned to that cluster, and their distances to that clustering center are calculated. If a pixel's distance to multiple clustering centers is less than a threshold T, it will be assigned to the nearest cluster. This work employs a similar grid search method, where different threshold values are tested. The performance is assessed using clustering evaluation metrics such as the silhouette score or Davies–Bouldin index. The threshold T that yields the best clustering performance on the validation set is selected.
3) For each superpixel block, the average position and color value of all pixels in it are calculated, and its center point is updated. The process is repeated until convergence.

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Note that while auxiliary information has not been considered in the original method, it can be integrated into the SLIC method to enhance the accuracy of the superpixel segmentation results. To achieve this, the distance metric used in the original SLIC is combined with auxiliary data. Specifically, the distances computed for the segmentation now include

$$\begin{cases} d_c = \sqrt{\left(a_j - a_i\right)^2 + \left(b_j - b_i\right)^2} \\ d_s = \sqrt{\left(x_j - x_i\right)^2 + \left(y_j - y_i\right)^2} \end{cases} \tag{1}$$

where (a_i, b_i) and (a_j, b_j) represent the spectral features of the center pixel and its neighboring pixel, respectively, and (x_i, y_i) and (x_j, y_j) denote their corresponding spatial coordinates. The term d_c represents pixel spectral differences, and d_s represents the spatial distance, each computed in a 3×3 pixels window

$$D = \sqrt{\lambda\left(\frac{d_c}{N_c}\right)^2 + \left(\frac{d_s}{N_s}\right)^2} \tag{2}$$

where N_c and N_s are normalization factors used to balance the magnitudes of spectral differences and spatial distances, respectively, and λ is a weighting parameter that adjusts the relative importance of spectral similarity versus spatial proximity. The parameter λ balances the contribution of two distances in our method. We use cross-validation to find the optimal λ. By dividing the dataset into training and validation subsets, we systematically vary λ and measure the performance based on the method's objective, such as classification accuracy or RE. The λ that maximizes performance on the validation set is considered optimal.

After superpixel segmentation, we employ SVM to classify each superpixel based on its spectral and spatial characteristics. For each superpixel s_i, we construct a feature vector that combines multiple information sources

$$\mathbf{f}_i = \left[\boldsymbol{\mu}_i, \boldsymbol{\delta}_i, \mathbf{p}_i\right] \tag{3}$$

where $\boldsymbol{\mu}_i$ represents the mean spectral signature of the superpixel, $\boldsymbol{\delta}_i$ captures the spatial relationship with OSM objects (such as distances to the nearest road or building), and $\mathbf{p}_i$ indicates the proportion of different OSM categories overlapping with the superpixel. The spatial relationship vector $\boldsymbol{\delta}_i$ is computed as a normalized vector of distances $[d_{\text{road}}, d_{\text{build}}]$, where d_{road} is the minimum Euclidean distance from the superpixel centroid to the nearest road segment and d_{build} is the minimum distance to the nearest building boundary. These distances are normalized by their respective maximum values in the scene to ensure comparable scales.

Using these feature vectors, we train an SVM classifier with RBF kernel to determine the category of each superpixel

$$c_i = \text{SVM}(\mathbf{f}_i), \quad c_i \in \{\text{water, building, road, other}\}. \tag{4}$$

The training samples are selected from superpixels with high confidence based on their overlap with OSM objects. Here, "high confidence" means that a superpixel has more than 80% of its area overlapping with a single type of OSM object (water body, building, road, or other land use type). This strict threshold ensures that the selected training samples are representative of their assigned classes with minimal ambiguity at their boundaries. For each class (water, building, road, other), we randomly selected 70 superpixels with over 80% overlap with the corresponding OSM objects as training samples, and another 30 superpixels per class as validation samples to tune the SVM parameters. This sampling strategy ensures a balanced representation of each class while maintaining high confidence in the ground-truth labels. This classification strategy effectively combines the spectral information from hyperspectral data with the spatial context from OSM data, enabling robust categorization of urban land-cover types.

B. Mining Similar Information Based on Graph Associations

The primary objective of this section is to describe the method of utilizing graph associations to identify similar blocks in hyperspectral images. In remote sensing image analysis, it is common to encounter images where similar blocks exhibit significant scale differences. These differences may arise from various factors, including altitude variations, sensor resolution differences, or the sizes of objects. To effectively capture these cross-scale similarities and enhance search efficiency, the graph construction method presented here employs a cross-scale graph association construction approach. This technique enables the exploration of similarities across different scales by dynamically constructing a graph that links similar patches at varying resolutions. Such an approach allows for the identification of more similar blocks, thereby improving the accuracy of computer vision tasks like object detection and recognition. Overall, the implementation of cross-scale graph association construction methods marks a significant advancement in the field of remote sensing image analysis, offering substantial potential for future research and application development.

Cross-scale graph associations construction methods [54] generally comprise two primary steps: initially, the search for similar blocks, followed by graph construction. In the first step, algorithms are employed to identify blocks that closely resemble a given block. These identified blocks are then utilized to construct a graph, where each block represents a node, and the edges signify the similarity between nodes at varying scales. This method facilitates the capture of cross-scale similarities, allowing for a more comprehensive understanding of the images.

The method for obtaining feature vectors is based on cross-scale spatial similarity. Let $F^{\uparrow}$ and $F^{\downarrow}$ denote the high-resolution and low-resolution images, respectively. For each target block (black square) in $F^{\uparrow}$, similar blocks (yellow box) are sought in $F^{\downarrow}$. Since $F^{\downarrow}$ is derived by downsampling $F^{\uparrow}$, corresponding mapping blocks (red box) in the high-resolution image $F^{\uparrow}$ can be identified. These block pairs capture the intrinsic relationship between images at different resolutions, effectively leveraging the cross-scale recurrence property. The feature vector construction process can be formulated as

$$\mathbf{v}_i = \mathbf{s}_i, \mathbf{p}_i, \mathbf{r}_i \tag{5}$$

where $\mathbf{v}_i$ represents the feature vector of the ith block pair, $\mathbf{s}_i$ denotes the spatial information including the coordinates

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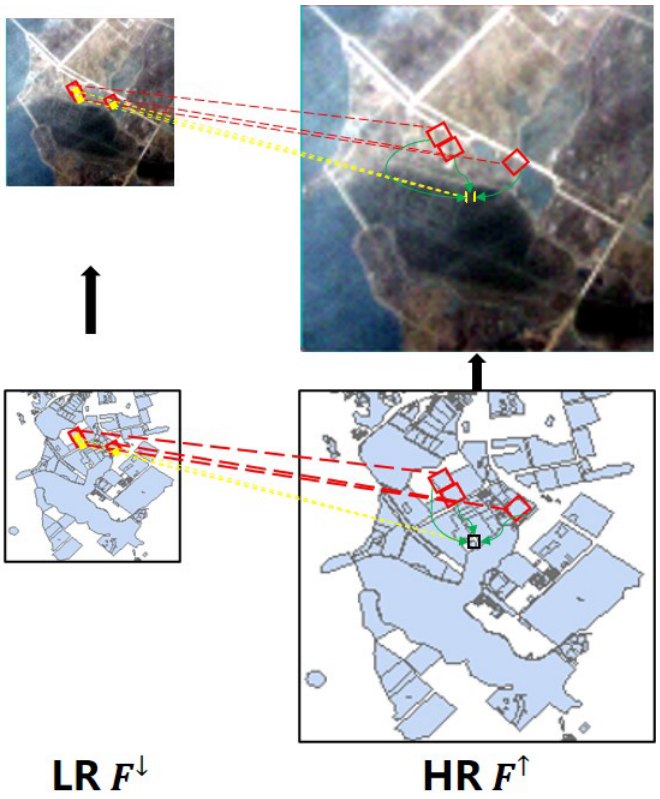


Fig. 2. Flowchart for obtaining similar information by the graph.

of blocks in the image, $\mathbf{p}_i$ represents the spectral features containing the spectral response values within the blocks, and $\mathbf{r}_i$ indicates the similarity measure describing the degree of similarity between block pairs. Specifically, $\mathbf{s}_i = (x_i, y_i)$ contains the centroid coordinates of the block, $\mathbf{p}_i$ is computed as the mean spectral signature across all pixels in the block, and $\mathbf{r}_i$ is derived from the normalized similarity measures between the block and its corresponding blocks in other scales. The final feature vector is constructed as $\mathbf{v}_i = [\mathbf{s}_i^{\mathrm{T}}, \mathbf{p}_i^{\mathrm{T}}, \mathbf{r}_i^{\mathrm{T}}]^{\mathrm{T}}$, where each component is first normalized to [0, 1] to ensure comparable scales before concatenation.

Due to the low spatial resolution of hyperspectral images, directly finding similar blocks is challenging. As shown in Fig. 2, we employ an indirect method, involving the construction of graph associations based on auxiliary data, which is then mapped back onto the hyperspectral images. We calculate the similarity between blocks by measuring the Euclidean distance or cosine similarity between their feature vectors. Blocks with feature vectors that have high similarity (i.e., small Euclidean distance or high cosine similarity) are grouped together as similar blocks. Based on these findings, an undirected weighted graph is created. In this graph, the similar blocks serve as nodes, and the links between them are weighted according to both spatial distance and pattern similarity. Finally, these graph associations are projected back onto the original hyperspectral image. The unmixing results are then obtained by aggregating the outcomes for each target area with those of its corresponding similar blocks.

More specifically, graph construction is performed by a search using k nearest neighbors (NNs)

$$W = \lambda_1 W_D + \lambda_2 W_{\mathrm{SAM}} \tag{6}$$

where W_{SAM} represents the spectral similarity between two blocks. The distinction here is that while W_D measures the spatial distance between blocks, W_{SAM} specifically quantifies the similarity in their spectral characteristics. This ensures that the blocks are not only spatially close but also spectrally consistent, which is crucial for accurate unmixing in hyperspectral images. The combination of these two metrics (W_D and W_{SAM}) allows the algorithm to balance both spatial proximity and spectral similarity, leading to more robust and accurate results. λ_1 and λ_2 are the weight parameters used to measure the importance of the two influencing factors.

Therefore, W_D can be described as

$$W_D = \sqrt{\left(d_{\mathrm{target}} - d_{\mathrm{similar}}\right)^2}. \tag{7}$$

Here, W_D represents the spatial distance between similar blocks and target blocks. Specifically, d_{target} and d_{similar} are computed as the Euclidean distances between the spatial coordinates of the target block and its similar blocks, respectively. It should be noted that the selected approach is different from the usual visual processing graph associations construction methods. The goal of this article is to construct a graph associations for hyperspectral images. Therefore, spectral similarity is also an important influencing factor. In this article, spectral angle mapping (SAM) is exploited to compute spectral differences

$$\mathrm{SAM}(\mathbf{x}, \mathbf{y}) = \arccos\left(\frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}\right) \tag{8}$$

where $\mathbf{x}$ and $\mathbf{y}$ are the feature vectors being compared. This calculation is performed for both the target and similar blocks

$$W_{\mathrm{SAM}} = \left|\mathrm{SAM}_{\mathrm{target}} - \mathrm{SAM}_{\mathrm{similar}}\right|. \tag{9}$$

Here, $\mathrm{SAM}_{\mathrm{target}}$ and $\mathrm{SAM}_{\mathrm{similar}}$ are the SAM values for the target block and the similar block, respectively. Their combination provides a measure of the spectral similarity between the similar block's feature vector and the same reference or mean vector used for the target block. Finally, the graph associations that have been constructed are mapped onto hyperspectral images. This mapping aids in correcting potential errors during the unmixing process by utilizing similar blocks. The method capitalizes on the intrinsic correlation between similar blocks within hyperspectral images to enhance the accuracy of the unmixing process. This enhancement is achieved by constructing an irregular graph structure that encompasses all blocks as nodes. Nodes from different scales are interconnected with edges weighted by similarity, and this structure is utilized to aggregate information from high-resolution blocks. Not only does this approach improve the accuracy of the unmixing process, but it also offers benefits in terms of computational efficiency.

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C. Aggregated Final Results

This article uses an adaptive model allocation strategy that automatically determines the optimal unmixing model for each superpixel based on both local characteristics and global context. The allocation process can be formulated as

$$\text{Score}(s_i) = \alpha H(s_i) + \beta C(s_i) + \gamma S(s_i) + \eta G(s_i). \tag{10}$$

The Score in the above equation is designed to evaluate the suitability of different unmixing models for each superpixel based on multiple criteria, including homogeneity, spectral complexity, spatial context, and graph similarity. The rationale behind each component is as follows. The Score combines these factors using weighted terms (α, β, γ, and η), which are dynamically adjusted based on global scene statistics. This approach ensures that the unmixing model assigned to each superpixel is tailored to its specific characteristics, balancing accuracy and computational efficiency.

1) Homogeneity [$H(s_i)$]: Homogeneity measures the spectral uniformity in a superpixel. A higher homogeneity indicates that the superpixel is more likely to be well-represented by a simpler model (e.g., linear unmixing)

$$H(s_i) = 1 - \frac{\sigma(s_i)}{\max_j \sigma(s_j)} \tag{11}$$

where $\sigma(s_i)$ represents the standard deviation of spectral values in superpixel s_i.

2) Spectral Complexity [$C(s_i)$]: Spectral complexity quantifies the complexity of spectral mixing patterns in the superpixel. Higher complexity suggests the need for more sophisticated models (e.g., nonlinear unmixing)

$$C(s_i) = \frac{|\lambda_1 - \lambda_2|}{\lambda_1} \tag{12}$$

where λ_1 and λ_2 are the two largest eigenvalues of the covariance matrix of spectral signatures in s_i.

3) Spatial Context [$S(s_i)$]: Spatial context evaluates the influence of neighboring superpixels, ensuring that the model assignment is consistent with the spatial structure of the scene

$$S(s_i) = \frac{\sum_{j\in\mathcal{N}(i)} w_{ij} M(s_j)}{\sum_{j\in\mathcal{N}(i)} w_{ij}} \tag{13}$$

where $\mathcal{N}(i)$ represents the neighborhood of s_i, w_{ij} is the spatial weight based on distance, and $M(s_j)$ is the current model assignment of neighbor s_j.

4) Graph Similarity [$G(s_i)$]: Graph similarity incorporates the similarity between superpixels based on graph associations, ensuring that regions with similar characteristics are assigned similar models

$$G(s_i) = \frac{\sum_{j\in\mathcal{K}(i)} W(I_j, I_i) M(s_j)}{\sum_{j\in\mathcal{K}(i)} W(I_j, I_i)} \tag{14}$$

where $\mathcal{K}(i)$ represents the k-NNs of superpixel s_i in the graph associations and $W(I_j, I_i)$ is the graph weight defined in (6), combining both spatial distance (W_D) and spectral similarity (W_{SAM}).

The final model allocation is determined by

$$M(s_i) = \begin{cases} \text{LMM}, & \text{if Score}(s_i) < T_1 \\ \text{BMM}, & \text{if } T_1 \le \text{Score}(s_i) < T_2 \\ \text{NLMM}, & \text{if Score}(s_i) \ge T_2. \end{cases} \tag{15}$$

The thresholds T_1 and T_2 are adaptively determined using

$$\begin{aligned} T_1 &= \mu_s - \sigma_s \\ T_2 &= \mu_s + \sigma_s \end{aligned} \tag{16}$$

where μ_s and σ_s are the mean and standard deviation of scores across all superpixels, respectively.

Specifically, the classic linear AE model [19] is used for LMM. For BMM, we use DMBU [28], and for other nonlinear models, we use MAHUM [52].

Assuming the original hyperspectral image is Y, the solution can be described as

$$\mathbf{Y} = \sum_{c}^{C} \sum_{\text{method}}^{M} \psi_m^c \left(\mathbf{M}_m^c, \mathbf{a}_m^c\right) + n \tag{17}$$

where $c \in C\{\text{water, building, road, other}\}$ and $m \in C\{\text{LMM, BMM, NLMM}\}$. n is the residual value, which can be almost ignored. ψ_m^c represents different models.

When refining the unmixing results, we leverage the constructed graph to propagate and update abundance values through weighted averaging among similar blocks. This refinement process continues iteratively until convergence, ensuring spatial consistency while preserving spectral characteristics

$$\mathbf{Y} = \sum_{c}^{C} \sum_{m}^{M} \sum_{i}^{Q} W\left(I_{\text{similar}}, I_{\text{target}}\right) \psi_m^c \left(\mathbf{M}_m^c, \mathbf{a}_m^c\right). \tag{18}$$

I_{target} represents the target block's intensity or feature vector in the low-resolution image that we aim to enhance or process. I_{similar} denotes the intensity or feature vector of a block that is identified as similar to the target block. Similar blocks are determined based on feature similarity $\text{W}(\text{I}_{\text{similar}}, \text{I}_{\text{target}})$ computed through both SAM and WD as defined in [see (6)–(9)].

In our methodology, the variable Q denotes the number of similarity blocks, playing an essential role in data analysis and processing. Each similarity block encapsulates unique information, contributing collectively to a comprehensive understanding of the data. The selection of Q is influenced by the complexity and diversity of the dataset, allowing us to tailor the granularity and depth of our analysis by adjusting this parameter, thereby accommodating the specific traits of the dataset. The value of Q that provides the best balance between performance and computational cost is chosen. Furthermore, the function $W(,)$, crucial for computing aggregation weights for transformed neighbors, is central to our approach. Equation (6) formulates how the spatial distance and spectral similarity are balanced through weighting parameters λ_1 and λ_2 to compute the final aggregation weights. It considers various factors and interrelationships between data points, enabling a nuanced capture of each neighbor's contribution. Through meticulous design and optimization of

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this aggregation function, the model's efficacy in extracting pertinent insights, enhancing performance, and achieving heightened accuracy is significantly bolstered.

Algorithm 1 MAIN

Input: Hyperspectral image $\mathbf{Y}$, Raw auxiliary data from OSM

Output: Unmixing results for each pixel

1: Preprocess auxiliary data from OSM
   a. Remove redundant and invalid records
   b. Align OSM data with hyperspectral image coordinates
   c. Convert vector data to raster format
   d. Fill in missing values and correct topology errors
2: Implement superpixel segmentation and classification
   a. Perform SLIC segmentation with parameters S and T
   b. Calculate distances d_c and d_s using Equation (1)
   c. Compute combined distance D using Equation (2)
   d. Classify superpixels using SVM with preprocessed OSM data
3: Construct graph associations
   a. Extract feature vectors$\mathbf{v}_i$ using Equation (5)
   b. Calculate spatial distance W_D using Equation (7)
   c. Compute SAM using Equation (8)
   d. Calculate W_{SAM} using Equation (9)
   e. Build k-NN graph with weights W from Equation (6)
4: Calculate adaptive model allocation scores with graph information
   a. Compute homogeneity measure $H(s_i)$ using Equation (11)
   b. Calculate complexity measure $C(s_i)$ using Equation (12)
   c. Evaluate spatial context score $S(s_i)$ using Equation (13)
   d. Calculate graph similarity score $G(s_i)$ using Equation (14)
   e. Determine final score using Equation (10)
   f. Calculate adaptive thresholds using Equation (16)
   g. Assign unmixing models using Equation (15)
5: Implement multitask unmixing
   a. Apply appropriate unmixing model based on allocation results
   b. Calculate initial unmixing using Equation (17)
   c. Refine results using graph associations structure
6: Aggregate final results
   a. Calculate aggregation weights $W(I_{\text{similar}}, I_{\text{target}})$
   b. Combine unmixing results using Equation (18)
   c. Generate final abundance maps for each endmember

IV. Experiments

To verify the accuracy of the proposed algorithm, two novel datasets have been realized. Indeed, the use of urban auxiliary data requires suitable datasets, carefully designed to reflect real-world scenarios and the challenges faced in urban areas, such as complex building structures and varying lighting conditions. The results obtained using these datasets demonstrate the effectiveness of the proposed algorithm in handling these challenges and highlight its potential for practical applications in urban environments.

1) *Thomson Lake:* An airborne hyperspectral image about an area inside the city of Wuhan, China. The hyperspectral dataset consists of 133 bands with 100 × 100 pixels and covers nearly 9 km^2 with the spatial resolution 30 m. The area is characterized by a rather sparse urbanization, with small and narrow roads and buildings that are not very prominent.
2) *Pavia:* A hyperspectral image of the city of Pavia, Italy, acquired by the Italian Space Agency (ASI) using the PRISMA satellite. The dataset consists of 199 bands with 120 × 120 pixels and covers nearly 13 km^2 with the spatial resolution 30 m. This dataset is characterized by the fact that the river and channels are difficult to extract at the rather coarse spatial resolution of the spaceborne sensor. In addition, the area has a dense urbanization, with roads and buildings that are intertwined, a situation which makes it difficult to accurately extract a built-up map.

Regarding auxiliary information, the Thomson Lake dataset is used as an example. The urban auxiliary data is mainly composed of OSM data, supplemented by some social survey data. The social survey data includes land-use patterns from local planning offices, building occupancy information from census records, and transportation flow statistics from traffic monitoring systems. These social indicators help validate and enrich the OSM data, particularly in areas where physical features alone may not fully characterize the urban environment. For example, the building occupancy data helps distinguish between residential and commercial structures, while traffic flow information aids in understanding road utilization patterns. OSM is an open-source mapping project created and maintained by volunteers worldwide. It provides rich geographic information, including landmarks and facilities such as roads, buildings, and parks. In addition to OSM data, we conducted a series of data processing operations to make these data more suitable for our research purposes.

For the Pavia dataset, the auxiliary data includes comprehensive urban information obtained from multiple authoritative sources. The primary source is the Municipality of Pavia's geodatabase, which provides detailed urban planning data including building footprints, infrastructure networks, and land-use classifications. Similar to the Thomson Lake dataset, these data underwent thorough preprocessing including geometric correction, attribute standardization, and topology verification to ensure precise alignment with the PRISMA imagery.

First, we cleaned and filtered the OSM data to remove irrelevant or inaccurate information. This process involves identifying duplicate records through spatial overlap analysis (>95% threshold), removing entries with invalid coordinates or outdated timestamps, and verifying the geometric validity of all features. For handling missing data, we employ NN interpolation for attributes and apply morphological operations to ensure data continuity, while maintaining topological consistency between adjacent features. Furthermore, to combine OSM data with social survey data, we performed data

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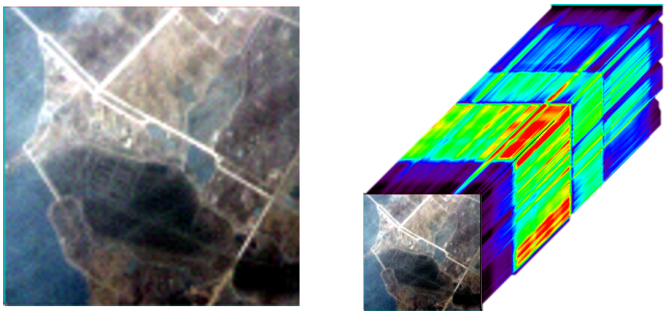


Fig. 3. True color representation and hyperspectral cube of the Thomson Lake test site.

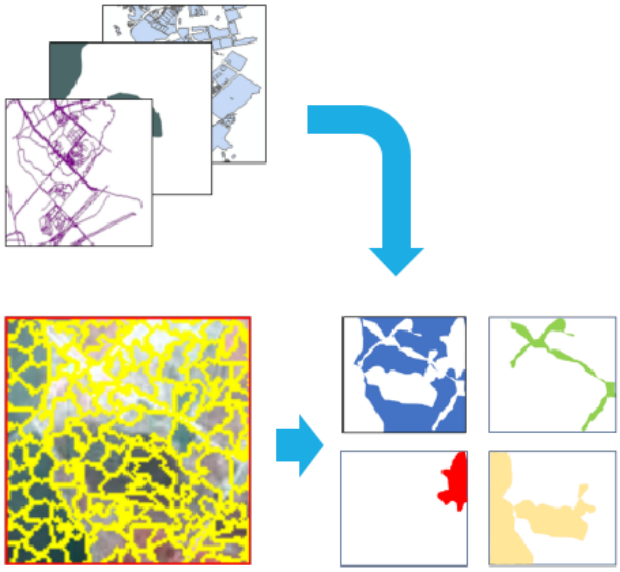


Fig. 4. Preprocessing of urban auxiliary data for the Thomson Lake test site.

matching and integration. This involved using geographic information system (GIS) technology to correspond to geographical elements from different data sources, establishing a consistent spatial reference framework for our research. We also executed data transformation and aggregation operations to create integrated datasets for analysis. Finally, to eliminate the impact of time intervals, we calibrated the data blocks.

Figs. 3 and 4 allow a visual comparison between the remotely sensed and the urban auxiliary data. The former provides detailed spectral information of the surface, while the latter better shows the layout and features of the city. By comparing these two types of data, a better understanding of the urban environment and its characteristics may be achieved.

Fig. 4 displays urban auxiliary data for the Thomson Lake test site. The preprocessing includes data cleaning, feature extraction, and normalization to ensure data quality and consistency. Each category of data is obtained through superpixel classification.

It is worth noting that preprocessing is an essential step in utilizing urban auxiliary data effectively. The raw data may contain errors or inconsistencies that can affect the accuracy of analysis results. Therefore, it is necessary to apply appropriate preprocessing techniques to ensure data quality and reliability.

A. Comparison With State-of-the-Art Methods

The effectiveness of our method through a comparative analysis involving a range of established unmixing algorithms. These encompass GBM [55], Semi-NMF [25], DMBU [28], 3DCNN [49], Semi-NMF-RE [41], and MAHUM [52].

It is worth noting that N-FINDR and VCA are primarily suited for endmember extraction, while algorithms like FCLS, GBM, and Semi-NMF are designed for determining abundance fractions. Consequently, a combined approach is required in the experiment. Furthermore, VCA, N-FINDR, FCLS, uDAS, $L_{1/2}$-NMF, and Semi-NMF-RE focus on linear unmixing, while GBM, Semi-NMF, rNMF, DMBU, 3DCNN, and MAHUM employ nonlinear unmixing techniques. The proposed method is the only one which incorporates auxiliary data, whereas the other algorithms rely solely on hyperspectral images.

It is essential to recognize that the RE serves as a supplementary evaluation metric for assessing the quality of unmixing results [25]. The RE measures the difference between the original hyperspectral image and its reconstruction, which can be formulated as

$$\mathrm{RE} = \frac{\|\mathbf{Y} - \hat{\mathbf{Y}}\|_F}{\|\mathbf{Y}\|_F} \tag{19}$$

where $\mathbf{Y}$ represents the original hyperspectral image and $\hat{\mathbf{Y}}$ denotes the reconstructed image. $\|\cdot\|_F$ indicates the Frobenius norm. The RE value ranges from 0 to 1, with smaller values indicating better reconstruction quality.

B. Results for the Thomson Lake Test Area

The distribution of different mixing patterns in the Thomson Lake dataset shows that 18.32% of the area exhibits linear mixing characteristics (LMM regions), mainly corresponding to homogeneous water bodies and open spaces. Complex nonlinear mixing (NLMM regions) accounts for 11.45% of the scene, primarily in dense urban areas with multiple scattering effects. The remaining 70.23% represents moderately complex regions (BMM regions), typically including residential areas and mixed vegetation zones. As shown in Table I, different unmixing methods demonstrate varying performance across different region types in the Thomson Lake dataset. Linear methods (N-FINDR + FCLS, VCA + FCLS, and L1/2-NMF) achieve their best performance in LMM regions, with Semi-NMF-RE reaching the lowest RE of 0.1096. In contrast, nonlinear methods like 3D-CNN show superior performance in NLMM regions (RE = 0.1118), demonstrating their advantage in handling complex nonlinear mixing patterns. Our method maintains stable performance across all regions, with REs of 0.1276, 0.1176, and 0.1076 for LMM, BMM, and NLMM regions, respectively, validating the effectiveness of our adaptive model selection strategy.

Looking at Fig. 5, the performance of GBM, Semi-NMF, DMBU, and 3DCNN, when deployed without auxiliary data,

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TABLE I
RE OF THE THOMSON LAKE TEST SITE

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	Semi-NMF-RE	MAHUM	Our
RE(LMM)	0.2814	0.2545	0.2512	0.2322	0.2391	0.2020	0.1918	0.1721	0.1518	**0.1096**	0.1487	0.1276
RE(BMM)	0.3114	0.2845	0.2812	0.2622	0.2191	0.1620	0.1518	0.1521	0.1318	0.1396	0.1387	**0.1176**
RE(NLMM)	0.3314	0.3045	0.3012	0.2822	0.1991	0.1920	0.1818	0.1321	**0.1118**	0.1496	0.1087	0.1076
RE(Thomson Lake)	0.3014	0.2745	0.2712	0.2522	0.2191	0.1820	0.1718	0.1521	0.1318	0.1296	0.1287	**0.1176**

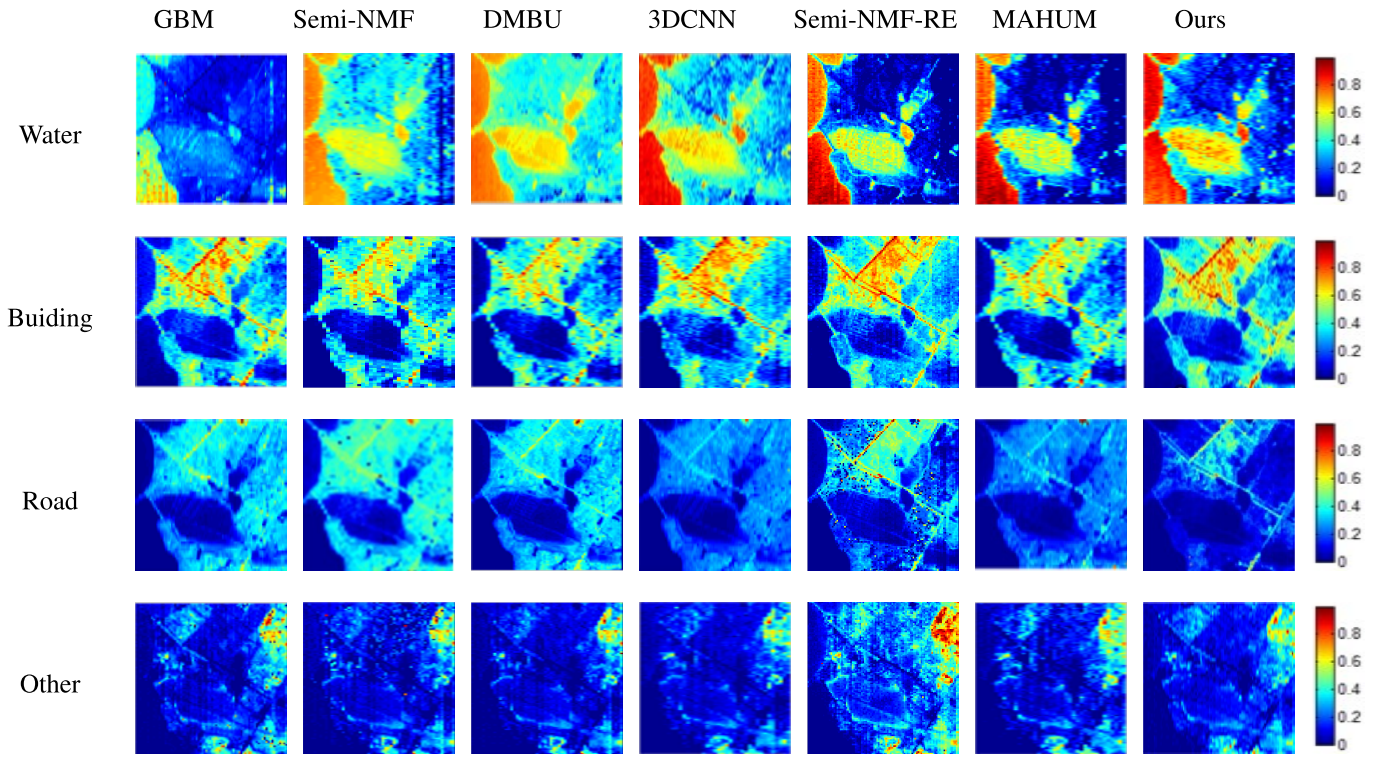


Fig. 5. Abundance maps of the water, buildings, roads, and others from the Thomson Lake dataset obtained by the different algorithms.

exhibits noticeable shortcomings. A striking number of buildings and roads are incorrectly classified as water bodies. The Semi-NMF-RE approach shows generally promising results; however, it suffers from significant misclassification issues in the other category. Furthermore, its capability in water detection is less effective compared to our proposed method. Although MAHUM's overall performance was fair, it was flawed in some details, such as the misclassification of roads. This serves as an empirical indicator that a dearth of additional contextual data can adversely impact the accuracy of SU.

The proposed technique, instead, incorporating auxiliary data, manages to evade such pitfalls. From the visual evidence provided, it is evident that the algorithm excels in the accurate demarcation of water areas. Moreover, the method's proficiency is not limited to water bodies alone; it also successfully identifies and categorizes buildings with a high degree of accuracy.

When it comes to roadways, the results exhibit a diverse range of features that align closely with real-world conditions. Unlike traditional methods, which may oversimplify complex terrains, our algorithm takes into account multiple factors that contribute to the heterogeneity of road surfaces. This could include variations in material, lighting, and other environmental factors, thereby ensuring a more accurate and nuanced interpretation of the scene.

In summary, not only does the proposed method perform well in the accurate identification of distinct elements like water and buildings but it also provides a more sophisticated, multidimensional representation of complex structures, such as roads. The addition of auxiliary data acts as a force multiplier, enabling our algorithm to surpass conventional techniques that rely solely on hyperspectral images for classification and unmixing.

C. Results of the Pavia Test Site

In the Pavia dataset, the distribution of mixing patterns differs significantly due to its dense urban characteristics. Only 5.24% of the area shows linear mixing (LMM regions), mainly in open squares and wide roads. Nonlinear mixing (NLMM regions) is more prevalent, accounting for 31.68% of the scene due to the complex historic architecture and narrow streets causing multiple reflections. The remaining 63.08% exhibits moderate mixing complexity (BMM regions), corresponding to areas with less dense urban structures. The quantitative results in Table II further support our visual analysis. In the Pavia dataset, linear methods perform particularly

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TABLE II
RE OF THE PAVIA TEST SITE

	N-FINDR+FCLS	VCA+FCLS	$L_{1/2}$-NMF	uDAS	VCA+GBM	VCA+Semi-NMF	rNMF	DMBU	3D-CNN	Semi-NMF-RE	MAHUM	Our
RE(LMM)	0.1859	0.1658	0.1355	0.1533	0.1893	0.1451	0.1528	0.0926	0.0919	**0.0610**	0.0925	0.0795
RE(BMM)	0.2159	0.1958	0.1655	0.1833	0.1693	0.1251	0.1328	0.0826	0.0819	0.0810	0.0825	**0.0695**
RE(NLMM)	0.2259	0.2058	0.1755	0.1933	0.1493	0.1451	0.1528	0.0726	**0.0619**	0.0910	0.0625	0.0595
RE(Pavia)	0.2059	0.1858	0.1555	0.1733	0.1693	0.1351	0.1428	0.0826	0.0719	0.0710	0.0725	**0.0685**

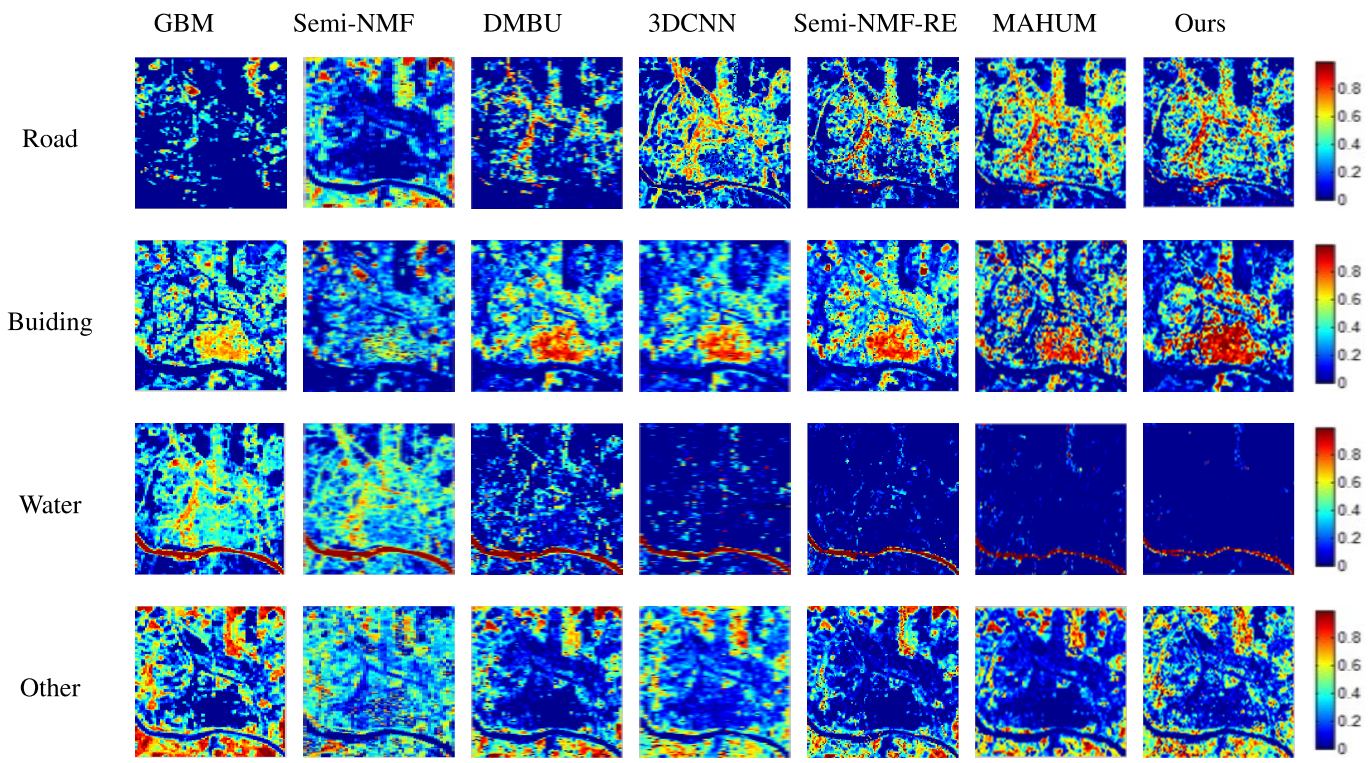


Fig. 6. Abundance maps of the water, buildings, roads, and others from the Pavia PRISMA dataset obtained by the different algorithms.

well in LMM regions, with Semi-NMF-RE achieving an RE of 0.0610. For complex urban areas (NLMM regions), nonlinear methods like 3D-CNN demonstrate better performance (RE = 0.0619). Our method achieves consistent performance across different region types (RE = 0.0795, 0.0695, and 0.0595 for LMM, BMM, and NLMM regions, respectively), showcasing the advantage of combining multiple models with auxiliary data guidance.

From the result presented in Fig. 6, it is clear that GBM, Semi-NMF, DMBU, and 3DCNN generally show suboptimal results. Specifically, these methods exhibit significant misclassification issues, particularly with water bodies. Semi-NMF-RE demonstrates notable deficiencies in water classification and lacks the precision in road detail extraction that our proposed method achieves. Although MAHUM shows some discrepancies, too, and imperfections in certain water regions, it performs relatively better. However, some noise was observed in the results for building areas. Moreover, many areas that are water are not recognized, signaling a clear deficiency in the algorithm's capabilities.

The shortcomings extend to other detection as well; a significant number of other patches are not effectively identified. This is concerning, as other forms an essential part of the terrain and is crucial for various applications, such as agriculture or land-use planning.

Furthermore, when it comes to the classification of more complex structures like scattered roads and buildings, MAHUM falls noticeably short. A large portion of buildings is wrongly classified as roads, indicating a general lack of precision in the algorithm's ability to distinguish between various urban features.

These observations collectively underscore the need for improvements to the MAHUM approach, especially when dealing with diverse and complex terrains. The algorithm appears to lack the sensitivity and specificity required for a nuanced understanding of different land features.

From the performance of the proposed method, we find that it excels in multiple aspects, particularly in the detection of buildings. The approach is capable of extracting more accurately defined structures, thanks in large part to the utilization of auxiliary data. This additional information aids in achieving more refined superpixel segmentation during the experimental phase.

The complexity of the scene poses another challenge, as this dataset contains more than just the four initially identified endmembers. Despite this, our method effectively mitigates the

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TABLE III
COMPUTATIONAL TIME ANALYSIS (IN SECONDS)

Method	Thomson Lake	Pavia
GBM	659.7	682.3
Semi-NMF	826.5	845.8
DMBU	3093.8	3212.4
3DCNN	2766.6	2785.9
Semi-NMF-RE	857.9	878.2
MAHUM	1625.0	1648.6
Ours	985.2	1006.7

issue by employing a graph association. This advanced feature allows our algorithm to reduce noise and anomalies, thereby enhancing the accuracy of the endmember classifications.

Overall, the proposed method delivers promising outcomes in hyperspectral unmixing and exhibits a strong capacity to manage complex terrains and multiple reflections. We are optimistic that the effectiveness of our approach can be further elevated by integrating more advanced techniques and exploring innovative avenues for the application of auxiliary data.

D. Computational Costs

The testing of this algorithm was carried out on a computer with the following specifications: Intel[1] Core[2] i7-8700 K CPU (3.70 GHz), 32 GB RAM, and a 3090 GPU. Table III shows the execution times under identical experimental settings. While our method is not the fastest among all compared approaches, it achieves a good balance between computational efficiency and unmixing accuracy. Compared to advanced deep learning-based methods like DMBU (3093.8 s) and 3DCNN (2766.6 s), our method (985.2 s) significantly reduces the computational time by about 68% and 64%, respectively, while maintaining superior unmixing accuracy. Although traditional methods like GBM (659.7 s) are faster, they show considerably lower accuracy as demonstrated in our unmixing results. This efficiency–accuracy tradeoff is consistent across both datasets, demonstrating the robustness of our approach.

V. CONCLUSION

This article provides information on a multitask AE hyperspectral unmixing model with information gain based on graph associations applied to auxiliary data. The model is designed to address the challenges posed by complex mixed terrain scenes, such as urban areas, by utilizing urban auxiliary data to enhance the information of hyperspectral images. The framework performs superpixel segmentation on complex urban environments and selects different AE unmixing methods based on the segmentation results. Additionally, graph associations are employed to retrieve similar blocks in the image and explore additional information present in hyperspectral images. The proposed method achieves good performance even in complex urban environments, as demonstrated by experiments conducted using two hyperspectral unmixing datasets with urban auxiliary data. Overall, this article provides valuable insights into improving hyperspectral unmixing accuracy in challenging environments using a multitask AE model and graph associations.

As for the future, this article mentions the use of multitask learning to solve the problem of hyperspectral image unmixing. However, the article does not mention how to select and design multiple tasks to maximize the performance of the model. In future research, it can be explored how to choose and design multiple tasks to better utilize the potential of the model. Additionally, the article does not mention how to handle the correlation and interaction between different bands, which may affect the performance and accuracy of the model. The article uses superpixel segmentation and graph associations to process hyperspectral images. However, these methods may increase the computational complexity, which may affect the real-time performance and scalability of the model. Additionally, more complex models or techniques can be considered to handle the correlation and interaction between different bands.

[1]Registered trademark.
[2]Trademarked.

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